Intelligent cache admission for LLM semantic caching: a literature survey

The research landscape for LLM semantic caching has exploded since 2023, with over 40 relevant papers spanning direct LLM response caching systems, learned admission/eviction policies, and foundational ML-for-caching work. GPTCache (2023) established the baseline architecture — embedding queries, performing ANN similarity search, and returning cached responses — but a rapid succession of follow-ups now addresses its core limitations: false-positive cache hits, context-insensitive matching, naive eviction policies, and static similarity thresholds. Meanwhile, a rich body of systems research on learned cache replacement (LRB, GL-Cache, HALP) provides the theoretical and practical foundations for designing intelligent Cache Value Functions that go beyond LRU/LFU. This survey organizes the most relevant papers across seven categories, covering 2020–2026 publications from top venues.

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1. Semantic caching systems purpose-built for LLMs

The foundational system in this space is **GPTCache**, introduced by Fu Bang (Zilliz) at the NLP-OSS Workshop co-located with EMNLP 2023. GPTCache converts user queries into embeddings via configurable models, performs similarity search through vector stores (arXiv) (FAISS, Milvus), and returns cached responses when cosine similarity exceeds a threshold. It supports modular eviction policies (LRU, LFU, FIFO) and multiple LLM adapters (OpenAI, LangChain). (GitHub) Reports **2–10× speed improvement** on cache hits. (ACL Anthology) Nearly every subsequent LLM caching paper uses GPTCache as a baseline. (arXiv)

MeanCache (Gill et al., Virginia Tech/Cisco, IPDPS 2025; arXiv:2403.02694) introduces user-centric caching with Federated Learning to collaboratively train an MPNet-based similarity model without sharing raw queries. It achieves **~17% higher F-score** and **~20% higher precision** than GPTCache, reduces storage by 83% via PCA embedding compression, (arXiv) and crucially introduces "context chains" for multi-turn conversation awareness (OpenReview) — preventing false hits where identical follow-up queries appear in different conversational contexts.

SCALM (Li et al., IWQoS 2024; arXiv:2406.00025) is the first paper to analyze real-world human-to-LLM interaction data for cache design. It proposes hierarchical semantic clustering to identify significant Q&A patterns, selective caching based on semantic pattern ranks, and dynamic eviction strategies. (arXiv) SCALM achieves a **63% increase in cache hit ratio** and **77% improvement in token savings** over GPTCache, (arXiv) (arXiv) and introduces "token saving ratio" as a new evaluation metric. (arXiv) (arXiv)

vCache (Schroeder et al., UC Berkeley/TU Munich, arXiv:2502.03771, February 2025) is the first verified semantic cache offering user-defined error rate guarantees. It observes that static similarity thresholds fail to provide formal correctness guarantees (arXiv) and proposes an online learning algorithm (VectorQ) that estimates **per-prompt optimal similarity thresholds** via

Bayesian sampling. (arXiv) Achieves up to **12.5× higher cache hit rates** and **26× lower error rates** compared to static-threshold baselines. (ResearchGate)

InstCache (Zou et al., arXiv:2411.13820, November 2024) takes a predictive approach: it uses LLMs to reorder the instruction representation space to develop spatial locality, then pre-populates the cache by predicting likely future instructions (arXiv) (Semantic Scholar) using negative log-likelihood scoring. (arXiv) Achieves **2.3× higher hit rate** than traditional caching (ResearchGate) on WildChat and reduces per-output-token time by up to 50% on vLLM.

GenerativeCache (Iyengar et al., arXiv:2503.17603, March 2025) goes beyond standard similarity matching to synthesize multiple cached responses into answers for queries never seen before. It adaptively varies semantic similarity parameters for different content types and includes a cost-aware model selection component. (arXiv) Claims **~9× faster** than GPTCache.

Several additional LLM caching papers deserve attention:

- **Towards Optimal Caching and Model Selection for Large Model Inference** (Zhu et al., NeurIPS 2023; arXiv:2306.02003) — jointly optimizes caching and model multiplexing with theoretically optimal algorithms for offline and online settings; (NeurIPS) uses Greedy Dual Size with Frequency replacement
- **GPT Semantic Cache** (Regmi & Pun, arXiv:2411.05276, November 2024) — Redis-based implementation reducing API calls by **up to 68.8%** (arXiv) (arXiv) with positive hit accuracy exceeding 97% (arXiv) (arXiv)
- **IC-Cache** (Yu et al., SOSP 2025; arXiv:2501.12689) — uses cached historical request-response pairs as in-context examples for smaller models rather than directly returning cached responses; (arXiv) improves throughput by **1.4–5.9×** (arXiv) (ADS)
- **SISO** (Kim et al., arXiv:2508.18736, 2025) — centroid-based semantic caching that clusters query vectors and caches centroids rather than individual vectors; (arXiv) achieves same hit ratios with **3× less memory** (arXiv)
- **GenCache** (Gill et al., arXiv:2511.17565, November 2025) — clusters structurally similar prompt-response pairs and generates programs to synthesize variation-aware responses; **83% cache hit rate** on Webshop (arXiv)
- **Semantic Caching for Low-Cost LLM Serving** (Liu et al., arXiv:2508.07675, August 2025) — proves optimal semantic cache computation is NP-hard; uses combinatorial multi-armed bandit framework for provably efficient online eviction (arXiv) (arXiv)
- **From Exact Hits to Close Enough** (Biton & Friedman, arXiv:2603.03301, February 2026) — systematic study of replacement policies for semantic caching; introduces SphereLFU, an online kernel density estimator that retains "prototypes" at medoids of high-density semantic regions (arXiv)
- **Cache me if you Can (OCaTS)** (Stogiannidis et al., Findings of EMNLP 2023; arXiv:2310.13395) — caches LLM responses to train a local student model (k-NN or MLP),

with criteria for when to trust the local model vs. calling the LLM; arXiv reduces teacher calls by ~1/3

- **waLLMartCache** (Dasgupta et al., ICPR 2025) — enterprise-scale distributed semantic caching with multi-tenancy support
- **LangCache-Embed** (Gill et al., arXiv:2504.02268, April 2025) — fine-tunes compact ModernBERT (~149M params) on domain-specific synthetic data, outperforming much larger embedding models for cache similarity judgment arXiv
- **An Ensemble Embedding Approach** (arXiv:2507.07061, 2025) — combines multiple embedding models through a trained meta-encoder; achieves 92% hit ratio while maintaining 85% rejection accuracy arXiv
- **Krites** (arXiv:2602.13165, February 2026) — asynchronous verified caching using an off-path LLM judge to verify borderline cache hits in a "similarity grey zone"; arXiv increases static-origin serves by up to **3.9×** arXiv
- **From Similarity to Vulnerability** (arXiv:2601.23088, January 2026) — first security analysis demonstrating key collision attacks on semantic caches arXiv

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2. Learned and intelligent cache admission policies

The evolution from heuristic to learned admission policies provides the core algorithmic foundation for intelligent LLM cache value functions. These papers are not LLM-specific but their techniques transfer directly.

LeCaR (Vietri et al., HotStorage 2018) pioneered ML-based cache replacement by using reinforcement learning with regret minimization to dynamically balance between LRU and LFU policies. It maintains online weights for each policy and updates them based on eviction outcomes. LeCaR outperforms ARC Jason Liu by more than **18×** when cache sizes are small relative to the working set, Arizona State University demonstrating that even simple online learning substantially improves upon established heuristics.

CACHEUS (Rodriguez et al., FAST 2021) extends LeCaR by classifying workloads into four primitive types (LFU-friendly, LRU-friendly, scan, churn) and designing specialized expert policies for each: SR-LRU for scan resistance and CR-LFU for churn resistance. It uses adaptive reinforcement learning with **automatically tuned hyperparameters** (learning rate and discount rate Frontiers via gradient-based stochastic hill climbing). NSF PAGES Evaluated on 17,766 experiments across 329 workloads, CACHEUS Arizona State University is the most consistently performing algorithm across diverse workloads and cache sizes. USENIX

TinyLFU (Einziger & Friedman, Euromicro PDP 2014; ACM Transactions on Storage 2017; arXiv:1512.00727) uses a compact approximate counting sketch based on Count-Min Sketch to estimate access frequency with very low memory overhead. It serves as a frequency-based gatekeeper that only admits items whose estimated frequency exceeds the eviction candidate's. arXiv The **W-TinyLFU** variant (window + SLRU main cache) is the basis for the Caffeine library,

ACM Digital Library Java's most widely used caching implementation. This is the primary non-ML admission baseline.

AdaptSize (Berger et al., NSDI 2017) is the first adaptive, size-aware cache admission policy for CDNs. It uses a probabilistic admission function that decreases exponentially with object size, optimized via a novel Markov chain model. ACM Digital Library Achieves **30–48% higher object hit ratio** than Nginx. Cmu The size-aware admission concept is directly relevant to LLM caching where responses have highly variable token counts.

RL-Cache (Kirilin et al., IEEE JSAC 2020) applies model-free reinforcement learning to CDN cache admission using **8 features** including object size, multi-timescale recency, and multi-timescale frequency. After training, it processes batches with ~16μs per-request overhead. Adityasundarrajan Outperforms AdaptSize on production Akamai CDN traces, demonstrating that multi-feature RL can learn effective admission strategies without manual feature engineering.

CacheSack (Yang et al., USENIX ATC 2022) is Google's production admission algorithm for datacenter flash caches (Colossus). It partitions cache traffic into disjoint categories, analyzes observed cache benefit per subset, and formulates a **knapsack problem** to assign optimal admission policies to each subset. USENIX The cost-aware approach considers both disk IO and flash wearout jointly. Deployed at Google scale, achieving **7.7% improvement in total cost of ownership** and **17.8% reduction in flash wearout**. ACM Digital Library

CACA (ACM Multimedia 2019) proposes content-feature-aware cache admission that admits video objects based on **content features** (topic, author, genre) rather than just request patterns. Uses tree-structure reinforcement learning to mine critical features from a large feature space. Improves hit ratio by up to 15%. ACM Digital Library Directly demonstrates multi-feature, content-aware admission using non-traditional signals — highly analogous to scoring LLM cache entries by query domain, response quality, or user context.

ALPS (Shahout & Friedman, OPODIS 2023) proposes a lightweight ML-based cache management meta-policy that derives features from space-efficient sketches (Count-Min Sketch, HyperLogLog), making the ML process lightweight. Achieves up to **20% improvement** over LRU with resource requirements orders of magnitude lower than LRB or other heavy ML approaches.

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3. Cache replacement policies informed by ML and Belady approximation

These foundational papers on learning-based cache eviction directly inform Cache Value Function design for LLM systems.

LRB — Learning Relaxed Belady (Song et al., NSDI 2020) uses gradient-boosted decision trees to approximate Belady's MIN oracle for CDN caching. Its key innovations include the **Relaxed Belady algorithm** (which treats all objects not requested within a boundary period as equivalent eviction candidates) and the **Belady boundary** concept. Microsoft USENIX LRB uses features derived from past request timestamps to predict whether an object's next access exceeds the

boundary. Princeton CS Reduces WAN traffic by **5–24%** over production CDN designs. USENIX Prototype implemented within Apache Traffic Server. Microsoft +2 This is the most influential paper connecting ML to cache eviction.

GL-Cache — Group-Level Learning (Yang et al., FAST 2023) proposes a middle ground between per-object learning (LRB: accurate but slow at ~100K QPS) and workload-level learning (LeCaR: fast but coarse). GL-Cache clusters similar objects into groups and learns eviction decisions at the group level, accumulating more training signals while amortizing overhead. GitHub Compared to LRB, GL-Cache achieves **228× higher throughput** and **7% higher hit ratio** on average across 118 production traces. Cmu The group-level concept maps naturally to grouping semantically similar LLM queries.

HALP (Song et al., NSDI 2023) is a production-deployed ML eviction algorithm for **YouTube's CDN**. It combines a heuristic scoring rule (request rate + end-of-chunk features) with a neural network reward model trained via preference learning with automated feedback — conceptually analogous to RLHF. Running in YouTube CDN production since early 2022, HALP achieves **9.1% byte miss reduction** at only **1.8% CPU overhead**. USENIX Class Central The hybrid heuristic+ML approach and production deployment methodology are the most directly relevant blueprint for deploying learned caching in LLM systems.

Parrot (Liu et al., ICML 2020; arXiv:2006.16239) frames cache replacement as imitation learning, training an LSTM-based neural policy to approximate Belady's oracle. On Google Web Search, Parrot increases hit rates by **61% over LRU**. Proceedings of Machine Le... arXiv The paper also releases a Gym environment for cache replacement research. ACM Digital Library

Hawkeye (Jain & Lin, ISCA 2016) learns from Belady's optimal algorithm using the OPTgen reconstruction algorithm, which simulates what OPT would have done on past accesses. A PC-based binary predictor classifies cache lines as "cache-friendly" or "cache-averse." This seminal paper established the **"learn from the oracle" paradigm** that LRB, Glider, Parrot, and Mockingjay all build upon. Semantic Scholar

Additional relevant replacement policy papers:

- **Glider** (Shi et al., MICRO 2019) — trains an attention-based LSTM offline, then distills to an Integer SVM implementable in hardware; reduces LLC miss rates by 8.9% over LRU NSF PAGES
- **Mockingjay** (Jain & Lin, HPCA 2022) — advances Belady approximation with multiclass reuse distance prediction instead of binary classification; IEEE Xplore improves IPC over LRU by 15.2% NSF PAGES
- **S3-FIFO** (Yang et al., SOSP 2023) — three static FIFO queues with quick demotion of "one-hit wonders"; lower miss ratios than all 12 state-of-the-art algorithms on 6,594 traces while achieving **6× higher throughput** than optimized LRU GitHub
- **LHD — Least Hit Density** (Beckmann et al., NSDI 2018) — evicts the object with the lowest predicted hits-per-space-consumed; Semantic Scholar requires **8× less space** than LRU to

achieve the same hit rate; Semantic Scholar USENIX directly foundational for multi-signal value scoring

- **Hyperbolic Caching** (Blankstein et al., ATC 2017) — assigns priority = frequency / age with pluggable cost functions supporting cost-aware, expiration-aware, and cost-class caching; reduces miss rates by 10–20% ACM Digital Library
- **ARC** (Megiddo & Modha, FAST 2003) — adaptive replacement that dynamically balances recency and frequency via shadow caches; the gold standard adaptive heuristic baseline
- **OLeCaR** (Yusuf et al., arXiv:2009.11330, 2020) — formalizes cache replacement as a multi-armed bandit with delayed feedback and decaying costs; proves vanishing regret guarantees arXiv
- **LHR** (Yan et al., CoNEXT/IEEE ToN ~2022) — learns admission probabilities from the Hit Rate Optimal offline policy using XGBoost with 20 inter-request-time features NSF PAGES
- **RLR** (Sethumurugan et al., HPCA 2021) — reinforcement learning for hardware cache replacement implementable with 84μm² area overhead
- **HR-Cache** (Torabi et al., ICPE 2024; arXiv:2402.02795) — gradient-boosted decision trees based on hazard rate ordering for CDN edge caching arXiv

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4. Multi-signal cache value scoring and cost-aware caching

No existing paper explicitly combines frequency, cost, quality, and diversity into a unified LLM cache value function — this remains an open gap. However, several papers establish the components.

LHD (Beckmann et al., NSDI 2018) is the most principled approach, defining **"hit density" — expected hits per space consumed** — as a cache value metric. It uses conditional probability models incorporating object age, frequency, application ID, and size. DBS Bank USENIX This is the closest existing analog to a multi-signal Cache Value Function, and its hit-density concept (utility per space) transfers directly to LLM caching where entries consume variable token counts.

HALP (Song et al., NSDI 2023) demonstrates production-scale multi-signal scoring: a neural network trained via **preference learning** combines request rate, end-of-chunk features, and other signals into an eviction score. The preference-learning training methodology (analogous to RLHF) is particularly relevant for LLM cache admission where "correctness" of caching decisions can be evaluated post-hoc. Google Research

CacheSack (Yang et al., ATC 2022) formalizes cost-aware admission as a **knapsack optimization** problem, where each cache category has a measured benefit and cost (IO savings vs. flash wearout). This framework directly applies to LLM caching where different cached responses have different generation costs (varying model sizes, token counts, API prices). USENIX

RL for Adaptive Caching with Dynamic Storage Pricing (Sadeghi et al., IEEE JSAC 2019) formulates caching as an MDP with a value function combining popularity, storage costs, and fetching costs. Both model-based (Q-functions) and model-free RL approaches are developed. The value function naturally combines multiple signals into a unified optimization objective.

GD-Wheel and the GreedyDual family represent the classic approach to cost-aware replacement: items are scored by combining frequency and cost (e.g., recomputation time), and the lowest-priority item is evicted. (ResearchGate) GreedyDual-Size-Frequency (GDSF) is already used in the NeurIPS 2023 optimal caching paper (Zhu et al.) for LLM caching, where the "cost" is expected LLM processing time. (NeurIPS) The "From Exact Hits to Close Enough" paper (arXiv:2603.03301) also evaluates GDSF adapted for LLM workloads. (arXiv)

Cost-Aware Caching (Rossi et al., IEEE TPDS 2015) demonstrates caching that preferentially stores objects fetched through expensive external links, combining popularity with retrieval cost. This is analogous to preferentially caching responses from expensive models (GPT-4) over cheap models (GPT-3.5-Turbo).

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5. Semantic novelty and diversity as caching signals

Maximum Marginal Relevance (MMR) (Carbonell & Goldstein, SIGIR 1998) is the foundational algorithm for balancing relevance with information novelty. The MMR formula — score = $\lambda \cdot \mathrm{Sim}(di, Q) - (1-\lambda) \cdot \max(\mathrm{Sim}(di, dj))$ — penalizes redundancy against already-selected documents. MMR is now standard in RAG systems (LangChain, Pinecone, Chroma) and vector database retrieval, but has **not been formally applied to cache eviction decisions**. Adapting MMR to penalize redundant cache entries (entries whose semantic neighborhoods overlap heavily with other cached entries) is a clear research opportunity.

α-nDCG (Clarke et al., SIGIR 2008) extends NDCG to systematically reward novelty and penalize redundancy using subtopic coverage. It provides formal evaluation metrics for information diversity (ACM Digital Library) that could be adapted to measure — and optimize — cache content diversity.

PLMMR (SIGIR 2010) gives a principled probabilistic derivation of MMR via latent variable graphical models, removing the need to manually balance relevance and diversity parameters. This probabilistic formulation makes MMR more suitable for integration into formal cache value functions where the relevance-diversity tradeoff should be learned rather than hand-tuned.

PAMM (Xia et al., SIGIR 2015) uses policy gradient methods to directly optimize diversity evaluation measures within an MMR framework, demonstrating that MMR-style diversity can be learned end-to-end — relevant for learning cache diversity policies.

No paper was found that explicitly uses embedding-based novelty or diversity as a cache eviction signal. The SISO paper's centroid-based approach implicitly promotes coverage diversity by caching centroids that represent clusters, (arXiv) and SphereLFU (arXiv:2603.03301) retains "prototypes" at medoids of high-density semantic regions (arXiv) — both indirectly

related. **The explicit combination of MMR-style diversity scoring with cache admission/eviction is a significant gap in the literature.**

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6. Context-aware semantic caching for multi-turn conversations

ContextCache (Yan et al., PVLDB 2025; arXiv:2506.22791) is the most direct paper on context-aware multi-turn semantic caching. It employs a two-stage retrieval architecture: (1) vector-based retrieval on the current query to identify candidates, then (2) **self-attention mechanisms** to integrate current and historical dialogue representations for precise contextual matching. This overcomes attention dilution from simple concatenation and representation flattening from embedding averaging. Demonstrates improved precision and recall on ShareGPT conversations compared to GPTCache and MeanCache.

MeanCache's context chain mechanism (Gill et al., IPDPS 2025) records a "contextual chain" — the sequence of parent queries in the cache — for every cached entry. When a contextual follow-up arrives, MeanCache verifies its context by matching conversation history against cached context chains. (arXiv) GPTCache produces **54 false hits** on contextual queries versus MeanCache's **3 false hits** — a dramatic improvement in context sensitivity.

Context-based Semantic Caching for LLM Applications (IEEE, 2024; DOI:10.1109/10605418) proposes caching solutions that account for conversational context in cache key design, though detailed methodology is behind a paywall.

No paper named "LettuceCache" was found in any academic database or preprint server as of March 2026. The context-aware caching space remains relatively thin — most LLM semantic caching systems still treat queries as independent, single-turn inputs. Multi-turn context integration is identified as a key open problem across multiple papers.

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7. KNN/ANN-based cache lookup mechanisms

Approximate nearest neighbor search is the universal mechanism for semantic cache hit detection, but the specific design choices vary significantly across systems. **GPTCache** supports FAISS, Milvus, and Zilliz Cloud for ANN search with configurable embedding models. (GitHub) **SISO** introduces centroid-based caching — storing cluster centroids rather than individual query vectors — achieving the same hit ratios with **~3× less memory** (arXiv) and fundamentally changing the ANN lookup from nearest-individual-query to nearest-centroid matching.

vCache (arXiv:2502.03771) directly addresses the critical **threshold tuning problem** in ANN-based cache lookup. Rather than a single global cosine similarity threshold, vCache learns per-prompt-type optimal thresholds via online Bayesian sampling, recognizing that different query types require different similarity thresholds for accurate cache hit detection. (arXiv)

LangCache-Embed (arXiv:2504.02268) demonstrates that the choice of embedding model matters more than the ANN algorithm: domain-specific fine-tuning of compact ModernBERT

models outperforms much larger generic embedding models for similarity judgment in specialized domains like medicine. arXiv

RAGCache (Jin et al., ACM SIGCOMM 2024 / ACM ToCS; arXiv:2404.12457) integrates FAISS vector retrieval with a knowledge tree-organized KV cache and prefix-aware GDSF replacement, reducing TTFT by up to **4×** versus vLLM+FAISS. While focused on KV cache rather than response caching, its integration of ANN search with cache management is architecturally relevant.

The **Ensemble Embedding** paper (arXiv:2507.07061, 2025) shows that combining multiple embedding models through a meta-encoder achieves **92% cache hit ratio** while maintaining 85% rejection accuracy arXiv — suggesting that single-model ANN lookup may be fundamentally limited for semantic caching.

Conclusion

Three major findings emerge from this survey. First, **the LLM semantic caching field has matured rapidly but intelligent admission/eviction remains underdeveloped**. Most LLM caching systems (GPTCache, MeanCache, SCALM) still use simple LRU/LFU eviction arXiv despite the rich body of learned replacement work (LRB, GL-Cache, HALP) that could be adapted. Only a few very recent papers (arXiv:2508.07675, arXiv:2603.03301) begin to bridge this gap.

Second, **the explicit combination of multi-signal scoring with semantic diversity for cache admission is an open research gap**. No paper unifies frequency, generation cost, response quality, and MMR-style novelty into a single cache value function. The individual components exist — LHD and HALP for multi-signal scoring, MMR and α-nDCG for diversity, CacheSack for cost-aware optimization — but their integration into LLM caching is unexplored.

Third, **context-aware caching for multi-turn conversations is severely under-studied** relative to its practical importance. Only ContextCache (PVLDB 2025) and MeanCache's context chains address this directly, despite the fact that most real-world LLM usage involves multi-turn dialogue where identical queries can require completely different responses depending on conversation history. The production-deployed systems from CDN research (HALP at YouTube, CacheSack at Google) provide the clearest engineering templates for deploying intelligent caching at scale, but adapting their feature engineering and training pipelines to semantic embedding spaces — where "cost" includes LLM token generation time, "quality" includes response correctness, and "frequency" operates in continuous semantic space rather than discrete key space — remains an important open problem.